Received: 29 June 2023 | Accepted: 9 May 2024
DOI: 10.1111/eip.13562

ORIGINAL ARTICLE

WILEY

Exploring changing trends in depression and anxiety among adolescents from 2012 to 2019: Insights from My World repeated cross-sectional surveys

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Funding information
Jigsaw-The National Centre for Youth Mental Health, Grant/Award Number: CHY17439; ESB Energy for Generation Fund

Abstract

Aim: Research has indicated a rise in the prevalence of depression and anxiety among adolescents over the past three decades. However, the factors underpinning increases in mental health difficulties remain poorly understood. This study examines psychological, social and environmental risk and protective factors that may explain changes in depression and anxiety among adolescents.

Methods: Data were taken from two nationally representative My World Surveys of adolescents aged 12–19 years in 2012 ($N = 5{,}490$) and 2019 ($N = 9{,}844$). Survey data on depression and anxiety and a range of potential risk (e.g., alcohol use, psychotic symptoms) and protective factors (e.g., resilience, self-esteem) were assessed at both time points. Multiple group analyses assessed whether the predictive ability of risk/protective factors changed from wave 1 to wave 2.

Results: Results showed that the prevalence of depression and anxiety increased significantly between 2012 and 2019, particularly among females. Predictors accounted for between 37% and 61% of the variance in outcomes across waves. While some risk/protective factors were consistent predictors of depression and anxiety at both waves (e.g., bullying, discrimination, optimism), reporting female gender and having higher formal help-seeking tendencies more strongly predicted anxiety at wave 2, while lower self-esteem and lower resilience (personal competence) strongly predicted both depression and anxiety at wave 2.

Conclusion: Findings highlight the need to prioritize adolescent mental health service provision, especially in females. Self-esteem and resilience are potentially important targets for supporting adolescent mental health. Further research is required to understand the causal factors associated with increases in anxiety and depression.

KEYWORDS
adolescent, anxiety, depression, protective factors, risk factors

1 | INTRODUCTION

Adolescence is a critical period of development, which can shape the likelihood, severity and course of mental health problems (Kessler et al., 2007; Kessler et al., 2012). There is evidence that 62.5% of mental disorders occur before the age of 25 years (Solmi et al., 2022). However, mental health problems among this age group often remain undetected until adult life and many young people do not get adequate support (Hetrick et al., 2018). There is evidence indicating that the prevalence of mental health problems, particularly anxiety and depression, are increasing among adolescents (Bor et al., 2014; Collishaw, 2015). Although evidence from low-income countries is limited, studies from high-income countries including the United States (US; Keyes et al., 2019; Lu, 2019), New Zealand (Fleming et al., 2014), the United Kingdom (UK; Patalay & Gage, 2019; Sweeting et al., 2010) and other European countries (Thorisdottir et al., 2017; Torikka et al., 2014; Tørmoen et al., 2020; Von Soest et al., 2014) have pointed to increases in anxiety and depressive symptoms among adolescents, although rates vary across countries (Pitchforth et al., 2019). At the same time, some studies have reported that symptoms of depression and anxiety have plateaued or even decreased among adolescents (Ottová-Jordan et al., 2015).

Mental health difficulties are considered the leading contributor to the global burden of disease among young people and affect their potential to live a healthy life (Patel et al., 2007). Although trends in depression and anxiety have received research attention, the factors underpinning these trends remain inadequately understood (Cosma et al., 2020). Several factors have been proposed as contributing to increasing mental health problems in young people. These include social and economic changes related to family structure such as extended dependence on parents and delayed autonomy (Patton et al., 2016), parental conflict, and parent mental health difficulties (Sweeting et al., 2010). Other factors include: increased screen time, internet, and social media use (Carli et al., 2014; O'Keeffe et al., 2011); cyber bullying (Bottino et al., 2015); greater school-based pressures (Sweeting et al., 2010); changes in health (e.g., poorer sleep; Keyes et al., 2015); more sedentary lifestyles (Felez-Nobrega et al., 2020); cannabis use, eating problems, and appearance concerns (Von Soest et al., 2014).

Research indicates that mental health difficulties in adolescent girls have increased at a greater rate than boys (Potrebny et al., 2017), which has been attributed to a greater accumulation of worries about educational success, changing societal expectations of women and weight/appearance issues, which are thought to more negatively affect girls (Potrebny et al., 2017; Von Soest et al., 2014). Additionally, some have attributed the reported increases to greater social awareness of mental health difficulties, increased emphasis on mental health promotion in schools, and reduced stigma in help-seeking for mental health problems (Collishaw, 2015).

However, the literature lacks studies examining potential predictors of changes in mental health difficulties among adolescents (Von Soest et al., 2014). Additionally, there are a lack of studies on trends in mental health difficulties where identical procedures or measures have been used (Von Soest et al., 2014). Further, less research has considered whether decreases in potential protective factors, such as optimism, resilience, life satisfaction may result in increased mental health difficulties.

This study examines whether self-reported symptoms of depression and anxiety have increased between 2012 and 2019 in Ireland, using data from adolescents who participated in the first and second waves of the My World Survey (MWS-1; Dooley & Fitzgerald, 2012; MWS-2; Dooley et al., 2019). A repeated cross-sectional design is a valuable methodology when tracking self-reported mental health problems over time (Firebaugh, 2010). Few studies have examined the collective impact of individual, family, school, and social factors on adolescents' mental health (Lu, 2019), despite evidence highlighting their role as potential risk and protective factors (Long et al., 2021). Thus, the current study provides a unique opportunity, using an ecological risk/protective factor model, to assess which risk and protective factors are potentially predicting changes in depression and anxiety, which may in turn guide intervention efforts (Keyes et al., 2019). The present study addresses three research questions:

1. To what extent are increases in depression and anxiety observed between 2012 and 2019?
2. To what extent are changes in risk and protective factors observed between 2012 and 2019?
3. To what extent do changes in risk and protective factors potentially explain trends in depression and anxiety?

2 | METHOD

2.1 | Sample

Data were derived from a subset of the MWS-1 and MWS-2 datasets, which comprised of a representative sample of adolescents enrolled in post-primary education in the Republic of Ireland. For MWS-1, a random sample of 171 post-primary schools were invited to participate and 72 schools agreed (42% response rate). Data collection took place between February and October 2011. Responses were collected from 6085 adolescents (45% school response rate). Participants who did not report gender ($n = 32$) or age ($n = 23$) were removed, giving a final sample size of $n = 6032$. For MWS-2, 175 post-primary schools were invited to participate, and 83 agreed (47% response rate). Data collection took place between October 2018 and May 2019 and yielded valid responses from 10023 adolescents (50% school response rate). Participants were excluded if they did not report their age or gender ($n = 179$), giving a final sample size of $n = 9884$.

2.2 | Procedure

Ethical approval for both study waves was granted by the authors' institution. An email was sent to principals in each school with study information. After confirming a willingness to participate in the study,

TABLE 1 Outcome measures and internal consistencies (Cronbach's alpha) for the current sample.

Outcome	Scale	Wave 1 α	Wave 2 α
Negative domains			
Depression & Anxiety	*Depression, anxiety and stress scale* (DASS-21; Lovibond & Lovibond, 1995) Depression subscale (7 items), response range (0–42). Anxiety subscale (7 items), response range (0–42). 4-point Likert Scale (0–3). Total subscale score is multiplied by 2 to give subscale response range (0–42). Higher scores indicate greater levels of anxiety and depression over the previous week. *Standardized cut-offs (based on total subscale scores)* Normal range: 0–9 depression, 0–7 anxiety Mild–moderate: 10–20 depression, 8–19 anxiety, Severe/v severe: 21–42 depression, 15–42 anxiety	.88 .80	.91 .84
Psychotic symptoms	*Adolescent psychotic-like symptom screener* (APSS; Kelleher et al., 2011) 3-items, 3-point Likert scale Total response range (0–3). Higher scores indicate presence of more psychotic-like symptoms.	N/A	N/A
Alcohol use	*Alcohol use disorders identification test* (AUDIT; Saunders et al., 1993). 10 items, 5-point Likert scale (0–4). Total response range 0–40. Higher scores indicate more problematic alcohol use.	.82	.81
Mother & father criticism	*Network of relationships inventory: relationship qualities version* (NRI-RQV; Buhrmester & Furman, 2008) *Mother criticism*, 3 items, 5-point Likert scale (1–5), total subscale response range (3–15) *Father criticism*, 3 items, 5-point Likert scale (1–5), total subscale response range (3–15) Subscales score summed together, higher scores indicating higher parental criticism.	.84 .85	.85 .87
Avoidant coping	*Adapted coping strategy indicator* (CSI; Amirkhan, 1990) Avoidant coping subscale (6 items), 6-point Likert Scale (1–6). Total subscale response range 6–36. Higher scores indicate more avoidant coping.	.73	.75
Cannabis use	Participants indicated 1 ‘yes’ and 0 ‘no’ if they had ever smoked cannabis	N/A	N/A
Bullying & discrimination	Participants indicated 1 ‘yes’ and 0 ‘no’, whether they had ever been bullied and whether they had ever been treated unfairly because of their identity (e.g., sexual orientation, ethnicity, race, minority group status)?	N/A	N/A
Body dissatisfaction	Participants indicated how satisfied they were with their bodies using Likert-scale responses of 1 ‘very dissatisfied’ to 5 ‘very satisfied’.	N/A	N/A
Positive domains			
Self-esteem	*Rosenberg's self-esteem scale* (RSE; Rosenberg, 1965). 10 items, 4-point Likert scale (1–4), Total response range 10–40. Higher scores indicate higher levels of self-esteem.	.89	.84
Mother & father approval	*Network of relationships inventory: relationship qualities version* (NRI-RQV; Buhrmester & Furman, 2008) *Mother approval*, 3 items, 5-point Likert scale (1–5), total subscale response range (3–15) *Father approval*, 3 items, 5-point Likert scale (1–5), total subscale response range (3–15) Subscales score summed together, higher scores indicate higher levels of parental approval.	.82 .84	.83 .86

(Continues)

TABLE 1 (Continued)

Outcome	Scale	Wave 1 α	Wave 2 α
Optimism	*Life orientation test revised (LOT-R)*. (Carver & Schreier, 1985) 6 items, 5-point Likert scale (0–4), total response range (0–24) Higher scores indicate higher levels of dispositional optimism.	.89	.74
Life satisfaction	*Brief multidimensional students' life satisfaction scale—Peabody treatment progress battery* (BMSLSS-PTTB; Seligson et al., 2003) 6 items, 7-point Likert scale (1–7), total response range (6–42). Higher scores indicate higher levels of life satisfaction.	.81	.84
Resilience	*Resilience scale for adolescence* (READ) (Hjemdal et al., 2007) *Personal competence subscale* (self-efficacy and realistic orientation to daily life), 8 items, 5-point Likert Scale (1–5) total response range (8–40). *Family cohesion subscale* (supports within families, family's ability to maintain social outlook) 6 items, total response range (6–30). *Social competence* (social adeptness, communication skills) total response range, 5 items (5–25).	.77 .86 .74	.76 .72 .70
Peer and school connectedness	*Hemingway measure of adolescent connectedness* (MAC; Karcher & Lee, 2002) *Peer connectedness subscale* (6 items), 5-point Likert scale (1–5), Total response range (6–30). *School connectedness* (6 items), Total response range (6–30). Higher scores indicate higher connectedness.	.71 .82	.74 .80
Social support	*Multidimensional scale of perceived social support* (MSPSS; Zimmet et al., 1988) 12-item, 7-point Likert scale, (1–7), response range (12–84). Higher scores indicate higher levels of support. *Participants were asked to identify a special adult in their life and asked how regularly the person was available to them from irregularly to always (i.e., presence of one good adult).	.94	.94
Adaptive coping	*Adapted coping strategy indicator* (CSI; Amirkhan, 1990) *Problem focused subscale*, 5 items, 6-point Likert Scale (1–6). Total subscale response range 5–30. *Support focused subscale*, 4 items, 6-point Likert Scale (1–6). Total subscale response range 4–24. Higher scores indicate more problem focused and social support coping	.84 .77	.86 .76
Informal help-seeking	Participants were asked to indicate, 1 'Yes' or 0 'No' whether they talked about their problems to someone.	N/A	N/A
Formal help-seeking	Participants were asked if they had any serious problems (e.g., emotional, social, behavioural) in the past year that may have benefitted from professional help and whether they sought professional help or not for these problems.	N/A	N/A
Demographic characteristics & personal wellbeing			
	Participants indicated their age, gender, ethnicity and presence of a mental health difficulty and/or addiction in a parent.	N/A	N/A

all schools were sent information letters and consent forms for distribution to students and their parent/guardian(s). Upon return of the consent forms, a data collection date was set with each school; all schools were given the option of students completing either a paper or web-based version of the survey and to have data collection facilitated independently or with a research assistant.

2.3 | Measures

The measures employed are outlined in Table 1. Additional information is available in Dooley and Fitzgerald (2013) and Dooley et al. (2019).

2.4 | Data analysis

Descriptive statistics were conducted on sample demographics, independent variables ($n = 26$) and dependent variables (anxiety, depression) for waves 1 and 2 using SPSS (Version 26). One-way analyses of variance (ANOVA) evaluated differences in continuous variables across waves 1 and 2 while Chi-square tests of Independence, layered by wave, explored potential associations of wave on categorical variables. Chi-square statistics with values of $p < .01$ and standardized residuals $+/- 2$ were reported as significant (Agresti, 2007).

To test whether the stability of the associations between predictors and anxiety and depression varied across the two waves, a multi-group path analysis model was specified; the grouping variable was Wave 1/2. The scores on the anxiety and depression measures were simultaneously regressed on the predictor variables. The baseline model constrained all regression coefficients to be equal which represented a 'stability' model, with the associations between the predictors and anxiety and depression being the same at Wave 1 and Wave 2. Modification indices (MI) were used to indicate which constraints should be relaxed to improve model fit; that is, which regression coefficients were likely to be significantly different (Sörbom, 1989). MIs indicate which constraint could be relaxed so that fit of the model would significantly improve, that is, reduce the chi-square by 3.84 or more (the critical value for the chi-square for one degree of freedom, $p < .05$). A more conservative value of 5 was used to ensure that inconsequential parameters were not added and reduce the probability of type 1 errors (Murphy et al., 2019). A process followed whereby the equality constraint for the path with the largest MI was removed and the model was re-estimated. This continued until there were no MIs greater than 5. Overall model fit was assessed using the Comparative Fit Index (CFI ≥ 0.95), Tucker–Lewis index (TLI ≥ 0.90), the Root Mean Squared Error of Approximation (RMSEA ≤ 0.06) and Standard Root Mean Squared Residual (SRMR ≤ 0.08; Hu & Bentler, 1999). Chi-square test of model fit and Bayesian Information Criterion (BIC) were also assessed; a low Chi value relative to the degrees of freedom, higher p-value, and reductions in BIC indicated good model fit (Hu & Bentler, 1999). Robust maximum likelihood estimation was used to estimate the model parameters and handle missing data (<10%) and Mplus 6.1 was used (Muthén & Muthén, 1998–2017).

3 | RESULTS

Sample demographics for waves 1 and 2 are presented in Table 2. As Table A in S1, shows, adolescents in wave 2 were more likely to report greater severity of depression and anxiety. In 2012, 11.3% reported severe/very severe anxiety compared with 21.9% in 2019, while in 2012, 8% reported severe/very severe depression compared with 15.4% in 2019. Table B in S1, shows that females and older adolescents exhibited higher levels of anxiety and depression, but levels were significantly elevated in females in 2019 compared with 2012. Descriptive statistics for the variables included in the model at waves 1 and 2 are presented in Table 3. Table 4 presents bivariate correlations between predictor variables and outcome variables at waves 1 and 2.

The initial multi-group model with paths constrained to be equal across waves was an acceptable description of the data, χ^2 (48) $= 426.565$, $p < .001$, RMSEA $= 0.031$ (90% CI 0.029, 0.034), CFI $= 0.981$, TLI $= 0.961$, SRMR $= 0.013$. The sequence of inspecting MIs and relaxing the equality constraints resulted in nine paths being estimated separately for wave 1 and 2. The final model also demonstrated acceptable model fit. χ^2 (39) $= 58.421$, $p < .001$, RMSEA $= 0.008$ (90% CI 0.003, 0.012), CFI $= 0.999$, TLI $= 0.998$, SRMR $= 0.003$. Details of each modification are presented in Data S1.

The predictors explained a significant proportion of variance of the anxiety scores at Wave 1 ($R^2 = 0.369$) and Wave 2 ($R^2 = 0.473$), and the effects were larger for depression (W1 $R^2 = 0.487$, W2 $R^2 = 0.607$). As seen in Table 5, variables that positively predicted anxiety and depression included discrimination, bullying, psychotic-like experiences, formal help-seeking, avoidant coping, parental mental health/addiction problem and parental criticism. Variables that

TABLE 2 Sample characteristics for waves 1 and 2 of the my world survey.

	Wave 1 N (%)	Wave 2 N (%)
Gender		
Male	2936 (48.7%)	4369 (44.4%)
Female	3094 (51.3%)	5475 (55.6%)
Ethnicity		
White	5764 (96.5%)	8353 (86.8%)
Black	53 (0.9%)	259 (2.7%)
Asian	56 (0.9%)	244 (2.5%)
Other/mixed	99 (1.7%)	765 (8.0%)
	M (SD)	M (SD)
Age	14.93 (1.62)	14.79 (1.68)

TABLE 3 Descriptive statistics and comparisons of variables in model at waves 1 and 2.

	Wave 1 *N* (%)	Wave 2 *N* (%)	χ^2
Parental mental health problem/addiction	716 (13.3%)	1707 (17.0%)	37.81***
Cannabis use	712 (12.0%)	1455 (15.2%)	30.90***
Help-seeking: Few/no problems	3188 (53.5%)	3535 (37.7%)	443.83***
Help-seeking: Some problems, but professional help not needed	1826 (31.3%)	3312 (35.3%)	
Help-seeking: Professional help needed but not sought	530 (9.1%)	1600 (17.1%)	
Sought professional when needed	358 (6.1%)	925 (9.9%)	
Discrimination	610 (10.6%)	1205 (13.6%)	29.55***
Bullied	2529 (44.6%)	3412 (34.0%)	40.16***
Body image: Dissatisfied	1907 (31.5%)	3084 (30.8%)	69.40***
Body image: Neutral	1244 (20.5%)	2615 (26.1%)	
Body image: Satisfied	2911 (48.0%)	4324 (43.1%)	
	M (SD)	**M (SD)**	**F**
Depression	7.25 (8.58)	9.88 (10.44)	246.60***
Anxiety	6.37 (7.33)	9.41 (8.98)	449.12***
Alcohol	3.91 (5.81)	2.75 (4.88)	172.30***
Social support	62.29 (15.49)	64.46 (5.26)	82.36***
Self-esteem	28.67 (5.70)	26.46 (5.26)	562.39***
Psychotic like experiences	0.65 (0.85)	0.87 (0.91)	220.56***
Problem focused coping	16.21 (5.64)	17.06 (5.54)	3.43
Support focused coping	14.63 (5.46)	14.46 (5.28)	147.68***
Avoidant coping	15.56 (6.07)	16.86 (6.21)	147.68***
School connectedness	20.40 (4.40)	20.10 (4.65)	14.36***
Peer connectedness	21.74 (3.64)	21.98 (4.21)	11.00**
Optimism	13.83 (4.69)	12.84 (4.77)	153.37***
Resilience (personal competence)	29.07 (5.26)	28.00 (5.80)	129.09***
Informal help-seeking	2.53 (0.79)	2.45 (0.83)	31.39***
One good adult	2.55 (0.76)	2.63 (0.71)	50.07***
Parental criticism	10.82 (5.04)	10.51 (4.89)	11.79**
Parental approval	23.26 (5.60)	23.38 (5.50)	1.37

Note: SR ≥ ± 2 are bolded.
Abbreviations: SR, standardized residuals; W1, wave 1; W2, wave 2.
****p* < .001; ***p* < .01.

negatively predicted anxiety and depression included optimism, peer connectedness, personal competence, self-esteem, informal help-seeking and having One Good Adult. Body satisfaction was specifically negatively associated with depression but not anxiety, while social support was only weakly positively predictive of depression at Wave 1. Age was negatively predictive of anxiety, whilst school connectedness and parental approval were weakly positively associated with anxiety but not depression.

As seen in Table 5, most of the regression coefficients of anxiety and depression did not differ significantly for Wave 1 and 2. However, being female gender and exhibiting higher help-seeking behaviours, lower self-esteem, and lower resilience (personal competence) were all more strongly associated with anxiety at Wave 2 than at Wave 1, whereas alcohol use became less strongly associated with anxiety at Wave 2 than at Wave 1. In terms of predicting depression, alcohol use and social support were less predictive at Wave 2 than Wave 1, whereas self-esteem and resilience (personal competence) were more predictive at Wave 2.

4 | DISCUSSION

Using data from two waves of a large community-based sample, this study investigated trends in adolescent mental health and examined a range of risk and protective factors associated with changes in anxiety and depression levels. In line with growing rates of adolescent mental

TABLE 4 Bivariate correlations between predictor variables and anxiety and depression at waves 1 and 2.

	Wave 1		Wave 2	
Predictors	**Anxiety**	**Depression**	**Anxiety**	**Depression**
Age[a]	0.10**	0.19**	0.06**	0.16**
Gender[a]	0.10**	0.15**	0.17**	0.15**
Ethnicity[a]	0.05**	0.04**	0.07**	0.06**
Parent/guardian mental health problems/addiction[a]	0.16**	0.17**	0.23**	0.24**
Cannabis[a]	0.15**	0.15**	0.11**	0.15**
Help-seeking behaviours[a]	0.34**	0.42**	0.48**	0.53**
Discrimination[a]	0.13**	0.11**	0.20**	0.21**
Bullying[a]	0.13**	0.12**	0.29**	0.28**
Body satisfaction[a]	−0.23**	−0.33**	−0.36**	−0.44**
Alcohol use	0.25**	0.27**	0.15**	0.20**
Social support	−0.18**	−0.21**	−0.31**	−0.23**
Self-esteem	−0.43**	−0.60**	−0.58**	−0.72**
Psychotic-like experiences	0.43**	0.37**	0.45**	0.42**
Planned coping	−0.11**	−0.21**	−0.22**	−0.31**
Support focused coping	−0.06**	−0.10**	−0.10**	−0.17**
Avoidant coping	0.40**	0.47**	0.48**	0.53**
School connectedness	−0.25**	−0.35**	−0.30**	−0.40**
Peer connectedness	−0.28**	−0.33**	−0.34**	−0.40**
Optimism	−0.37**	−0.51**	−0.48**	−0.59**
Resilience (personal competence)	−0.31**	−0.46**	−0.48**	−0.59**
Informal help-seeking	−0.20**	−0.29**	−0.27**	−0.34**
One good adult	−0.14**	−0.16**	−0.15**	−0.21**
Parental criticism	0.30**	0.32**	0.32**	0.39**
Parental approval	−0.22**	−0.33**	−0.25**	−0.34**

Note: All other correlations are Pearson's *r* correlations.
[a]Spearman's correlation.
**$p < .01$.

health difficulties (Keyes et al., 2019; Lu, 2019; Tørmoen et al., 2020), the prevalence of anxiety and depression was higher in 2019 compared to 2012, particularly in females and older adolescents.

Promisingly, adolescents in wave 2 exhibited higher levels of problem-focused coping, social support, peer connectedness, parent approval and the presence of One Good Adult, in addition to lower levels of risk factors such as bullying, alcohol use and parental criticism compared with wave 1. These findings align with the Health Behaviour in School-aged Children report (Gavin et al., 2020), which observed reductions in bullying and alcohol use and improved communication with parents in 10–18 year olds between 1998 and 2018. However, adolescents in wave 2 also exhibited comparatively lower levels of protective factors such as self-esteem, personal resilience and optimism, and were more likely to report a psychotic episode, cannabis use, problems that needed professional help, and having parent with a mental health difficulty/addiction.

Although levels of risk and protective factors varied across waves, some factors were shown to be more influential in predicting rates of anxiety and depression at a given wave. Gender was significantly associated with depression and anxiety at both waves, but being female was an important predictor of anxiety at wave 2. This is consistent with findings that mental health difficulties have increased in adolescent girls at a greater rate than boys (Potrebny et al., 2017; Wiens et al., 2020) and underscores the importance of gender-specific and early interventions (Lu, 2019). Consistent with other studies, trends did not vary across other demographic characteristics (e.g., ethnicity, age) (Keyes et al., 2019; Von Soest et al., 2014).

Several protective factors were shown to be negatively associated with anxiety and depression at waves 1 and 2, including high levels of planned coping, peer connectedness, optimism, informal help-seeking and having One Good Adult (Dooley et al., 2015; Sterrett et al., 2011). Conversely, lower self-esteem and lower personal resilience were more strongly associated with anxiety and depression at wave 2. These findings suggest that lower levels of personal resilience and self-esteem may, in part, explain the higher rates of anxiety and depression in adolescents, or alternatively, that higher rates of anxiety and depression may have led to reductions in adolescents' self-esteem and resilience (Hjemdal et al., 2007). Although longitudinal studies

TABLE 5 Unstandardized regression estimates for anxiety and depression at waves 1 and 2.

Predictors	W1 anxiety	W2 anxiety	W1 depression	W2 depression
Age	−0.200***		0.072	
Gender	−0.336*	0.612***	−0.354**	
Ethnicity	−0.038		−0.016	
Parent/guardian mental health problems/addiction	0.920***		0.704***	
Cannabis	0.080		−0.044	
Help-seeking behaviours	0.905***	1.302***	1.255***	
Discrimination	1.140***		0.978***	
Bullying	0.567***		0.305*	
Body satisfaction	−0.138		−0.332***	
Alcohol use	0.136***	0.027	0.094***	0.032
Social support	0.003		0.019**	−0.005
Self-esteem	−0.191***	−0.382***	−0.421***	−0.718***
Psychotic-like experiences	1.984***		1.151***	
Planned coping	0.086***		0.093***	
Support focused coping	0.008		−0.014	
Avoidant coping	0.194***		0.218***	
School connectedness	0.058**		0.003	
Peer connectedness	−0.093***		−0.082***	
Optimism	−0.094***		−0.159***	
Resilience (personal competence)	−0.049	−0.149***	−0.105***	−0.192***
Informal help seeking	−0.246**		−0.575***	
One Good Adult	−0.270*		−0.465***	
Parental criticism	0.137***		0.146***	
Parental approval	0.052***		0.009	
R^2	0.369	0.473	0.487	0.607

Note: A single regression coefficient across waves 1 and 2 for a given dependent variable indicates that parameters were constrained to be equal across waves. However, where two separate regression coefficients are presented for waves 1 and 2, this indicates that parameters were released and the strength of the effect of a given independent variable on the dependent variable varies by wave. Also note for categorical variables, regression estimations are reflective of how data were coded (e.g. for gender, male is the referent category against which female scores are compared). Females' negative scores indicate a weaker relationship and positive scores indicate a stronger relationship with outcome variables across waves.
Abbreviations: R^2, proportion of variance explained by the model; W1, wave 1; W2, wave 2.

would be required to parse out directionality of these effects, efforts to support adolescents' mental health should seek to build self-esteem, resilience and capacity to manage challenging circumstances.

Having problems needing professional help were more strongly associated with anxiety at wave 2 than wave 1. As expected, greater formal help-seeking tendencies were associated with higher levels of anxiety which may reflect the need to access professional help in light of the increase in anxiety, but it is unclear why this relationship was not observed for depression.

Alcohol use was significantly associated with anxiety and depression at wave 1 but not at wave 2, in line with von Soest et al. (2014) who also reported diminished relationships between alcohol intoxication and depression over time. Contrasting with Von Soest et al. (2014), cannabis use did not predict changes in mental health outcomes over waves, despite adolescents reporting greater cannabis use at wave 2. Further research is required to explore these relationships.

Discrimination and bullying were risk factors that consistently strongly predicted depression and anxiety at both waves and represent continued important targets for intervention (Arslan et al., 2021; Vines et al., 2017). Having a parent with a mental health difficulty and experiencing high levels of parental criticism were significantly associated with outcomes, indicating the importance of holistic, family-based approaches for supporting adolescent mental health (Sterrett et al., 2011). Having One Good Adult in their lives was a strong negative predictor of depression and anxiety in young people at both waves. But contrasting with Sweeting et al. (2010), parent-related factors did not appear to explain changes in anxiety or depression. Similarly, although lower school connectedness significantly predicted anxiety (McGraw et al., 2008) and higher body dissatisfaction predicted depression at both waves, contrasting with other research (Von Soest et al., 2014), these factors did not predict changes in mental health in this study.

4.1 | Strengths and limitations

Key strengths of this study were that it contained a large, nationally representative random sample of adolescents and investigated a range of socioecological factors using the same standardized measures and data collection procedures across waves (Bor et al., 2014). However, it is possible that the trends observed may be partly explained by changed vulnerability to factors that were not investigated (Von Soest et al., 2014). Due to constraints in the number of predictors permitted in the model, we were unable to include data collected on additional factors such as socioeconomic status and experiences of stressful life events (Kenny et al., 2013). Additionally, while data on adolescents' sleep hygiene, social media use and health behaviours (e.g., levels of physical activity, muscle building, weight loss behaviours) were collected in wave 2, comparable data were not available in wave 1, which prohibited investigating the influence of these factors on secular trends in mental health (Felez-Nobrega et al., 2020; Keyes et al., 2015, 2019). Further research is required to investigate these factors.

Research indicates that the risk/protective factors influencing mental health outcomes vary by gender. For example, societal expectations of women and worries about educational success or body image/appearance are thought to impact girls to a greater extent than boys (Potrebny et al., 2017). Gender differences in risk/protective factors were outside the scope of the present study, but future research should examine potential gender differences in important risk/protective factors for mental health. It is also possible that increases in mental health difficulties observed reflect greater social acceptance and willingness to report mental health symptoms (Collishaw et al., 2004, 2012), although RCTs have shown that a greater willingness to disclose mental health problems did not lead to increased reporting of mental health symptoms (Jorm et al., 2020). Foulkes and Andrews (2023) propose testing of the prevalence inflation hypothesis, stating that efforts to raise awareness about mental health problems have led to an inadvertent increase in reported rates of mental health problems via improved recognition but also overinterpretation. Finally, another limitation concerns the cross-sectional nature of the study which limits our capacity to assess causality or temporality of findings.

4.2 | Conclusion

Anxiety and depression levels have increased in adolescents in Ireland from 2012 to 2019. While some risk/protective factors were consistent predictors of anxiety and depression at both waves, reporting a female gender, having higher formal help-seeking tendencies and exhibiting lower self-esteem and personal competence more strongly predicted anxiety and/or depression at wave 2. Findings suggest that females should be prioritized in terms of service provision and that self-esteem and resilience represent potentially important targets for supporting mental health. As it was not possible to establish causal directions between anxiety and depressive symptoms and risk/protective factors, further research is required to guide efforts to reduce mental health difficulties among adolescents.

ACKNOWLEDGEMENTS

The authors would like to acknowledge and thank all of the young people who took part in this study and thanks to their parents, teachers and other school staff for supporting this research. Additional thanks to our My World Survey research team at University College Dublin School of Psychology. They would also like to acknowledge funding received from Jigsaw: The National Centre for Youth Mental Health and the ESB Energy for Generation Fund. Open access funding provided by IReL.

FUNDING INFORMATION

This research was supported by Jigsaw, The National Centre for Youth Mental Health, Ireland and the ESB Energy for Generation Fund.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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How to cite this article: Fitzgerald, A., Mahon, C., Shevlin, M., Dooley, B., & Reilly, A. O. (2025). Exploring changing trends in depression and anxiety among adolescents from 2012 to 2019: Insights from My World repeated cross-sectional surveys. *Early Intervention in Psychiatry*, *19*(1), e13562. https://doi.org/10.1111/eip.13562